predators' only source of nourishment, and that the predators never become full and never stop hunting. Finally, they assumed that environmental conditions, such as weather or natural disasters, had no impact on the process. The effect of the genetic diversity of the predators and prey animals on their ability to survive was not taken into account.

When plotted on a graph, the predator population trails the rise and fall of the prey population, and is still rising as the prey population starts to decline. This explained D'Ancona's observation of the larger proportion of predators after the prey population had been allowed to boom by a reduction in fishing.

The relative fluctuations of the populations depends on the relative reproductive rates of the two species and the predation rate. For example, oscillations in the size of an ant population and that of an anteater are barely noticeable because they reproduce at such different rates. The oscillations in the populations of species that breed at similar rates, such as the Iberian lynx and rabbit, are much more pronounced.